

Sets, Functions, and Cardinality

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COMP2711 Discrete Mathematical Tools

Overview

Suppose every student in a class needs a locker.

- **Sets** specify the students and lockers.
- A **function** assigns each student to one locker.
- Special functions tell us whether lockers are shared or unused.

Functions will later let us compare the sizes of infinite sets.

1 Sets

1.1 Membership and set descriptions

A **set** is an unordered collection of distinct objects. The objects are its **elements**. We write

$$x \in A \quad \text{when } x \text{ is an element of } A, \quad x \notin A \quad \text{otherwise.}$$

Throughout this course,

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\},$$

and $\mathbb{R}$ is the set of real numbers: all points on the number line.

There are two common ways to describe a set.

Method	Description	Example
List all elements	write every element	$O = \{1, 3, 5, 7, 9\}$
Set builder	state a membership test	$O = \{n \in \mathbb{N} \mid n < 10 \text{ and } n \text{ is odd}\}$

Set-builder notation needs a domain. The expression

$$\{x \in D \mid P(x)\}$$

contains exactly the elements x in D for which $P(x)$ is true. It is the **truth set** of P over D . Restricted quantifiers use the same domain convention:

$$\forall x \in D P(x) \equiv \forall x (x \in D \rightarrow P(x)), \quad \exists x \in D P(x) \equiv \exists x (x \in D \wedge P(x)).$$

Sets may contain strings, numbers, or even other sets. For example,

$$\{\text{"Ada"}, \text{"Ben"}\}, \quad \{-2, 0, \tfrac{1}{2}, \sqrt{2}\}, \quad \{7, \text{"hello"}, \{1, 2\}\}.$$

1.2 Set equality

MUST UNDERSTAND A standard set is unordered and deduplicated.

Two sets are equal when they have exactly the same elements:

$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

Order and repetition do not matter:

$$\{1, 3, 5\} = \{5, 3, 1, 3\}.$$

1.3 Membership and subsets

The symbols $\in$ and $\subseteq$ have different roles. We say that B is a **subset** of A when every element of B also belongs to A :

$$B \subseteq A \iff \forall x (x \in B \rightarrow x \in A).$$

Now the equality test can also be written

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

A **proper subset** is a subset that is not equal to the full set:

$$B \subsetneq A \iff B \subseteq A \text{ and } B \neq A.$$

Thus, A contains at least one element not in B .

For $A = \{1, \{2, 3\}\}$,

$$1 \in A, \quad \{1\} \subseteq A, \quad \{2, 3\} \in A, \quad 2 \notin A.$$

The braces matter: 1 is a number, while $\{1\}$ is a set.

1.4 Set size

A finite set's **size**, also called its **cardinality**, is the number of distinct elements. It is written $|S|$. For example,

$$|\{a, b, c\}| = 3, \quad |\{1, 1, 2\}| = |\{1, 2\}| = 2.$$

The **empty set** is $\emptyset = \{\}$. It is the only set of size zero:

$$|\emptyset| = 0.$$

1.5 Power sets

The **power set** $\mathcal{P}(S)$ is the set of all subsets of S :

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}.$$

For $S = \{a, b\}$:

Choice bits	00	01	10	11
Subset	$\emptyset$	$\{b\}$	$\{a\}$	$\{a, b\}$

The first bit decides whether a is included; the second decides whether b is included.

If $|S| = n$, then

$$|\mathcal{P}(S)| = 2^n.$$

Why? A subset makes one independent yes/no choice for each element of S : include it or exclude it. Thus every subset corresponds to one bit string of length n .

1.6 Cartesian products

The **Cartesian product** of A and B is

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

For $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

If A and B are finite, then $|A \times B| = |A||B|$. More generally,

$$A_1 \times \cdots \times A_n = \{(a_1, \dots, a_n) \mid a_i \in A_i \text{ for every } i\}.$$

Each element is an **ordered pair**:

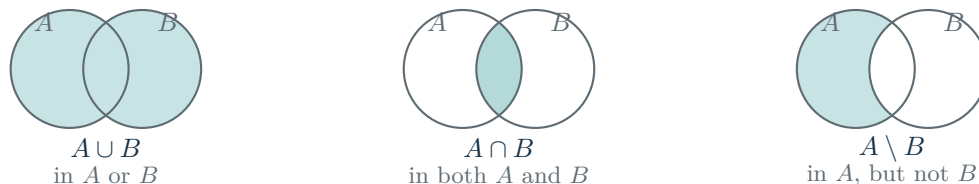
$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

2 Set operations

A **universal set** U contains all objects under discussion. The four basic operations are:

Operation	Set	Membership condition
Union	$A \cup B$	$x \in A$ or $x \in B$ (or both)
Intersection	$A \cap B$	$x \in A$ and $x \in B$
Difference	$A \setminus B$	$x \in A$ and $x \notin B$
Complement	$\bar{A} = U \setminus A$	$x \notin A$

A **Venn diagram** draws sets as regions. The shaded region shows the result.



Sets A and B are **disjoint** when $A \cap B = \emptyset$. For finite sets, the overlap must be subtracted once when a union is counted:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

2.1 Set identities

The membership conditions give a direct translation:

$$\cap \leftrightarrow \wedge, \quad \cup \leftrightarrow \vee, \quad x \in \overline{A} \leftrightarrow \neg(x \in A).$$

Therefore propositional equivalences produce set identities. For example,

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

is De Morgan's law.

Worked proof: an identity element by element

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Let x be arbitrary. Then

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff x \in A \wedge x \notin B \cup C \\ &\iff x \in A \wedge \neg(x \in B \vee x \in C) \\ &\iff x \in A \wedge x \notin B \wedge x \notin C \\ &\iff (x \in A \setminus B) \wedge (x \in A \setminus C) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

Because this holds for every x , the two sets are equal.

Other useful identities include

$$\begin{aligned} A \cup \emptyset &= A, & A \cap U &= A, \\ A \cup \overline{A} &= U, & A \cap \overline{A} &= \emptyset, \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C), & A \cap (B \cup C) &= (A \cap B) \cup (A \cap C). \end{aligned}$$

Rather than memorizing an isolated table, translate membership to logic when an identity is uncertain.

2.2 Families of sets

For a family $\{A_i\}_{i \in I}$,

$$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \in A_i)\}, \quad \bigcap_{i \in I} A_i = \{x \mid \forall i \in I (x \in A_i)\}.$$

Read the first as “the elements appearing in at least one set” and the second as “the elements appearing in every set.”

2.3 Computer representation

For $U = \{u_1, \dots, u_n\}$, a subset of U can be stored as n bits. Bit i is 1 exactly when u_i is in the subset. Then union, intersection, and complement become bitwise OR, AND, and NOT.

A **multiset** also records repetitions. Ordinary sets do not:

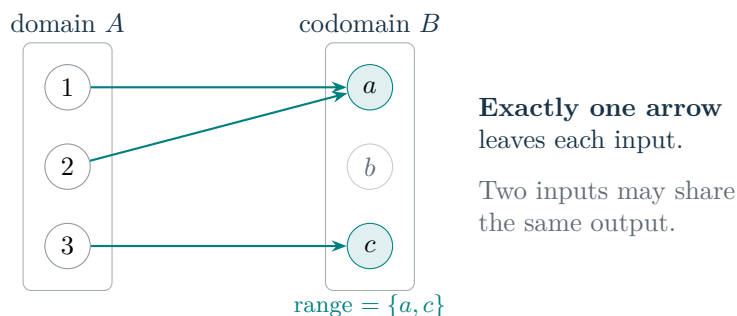
$$\{a, a, b\} = \{a, b\} \quad \text{as sets.}$$

3 Functions

3.1 Definition of a function

MUST UNDERSTAND Every input has exactly one output.

A **function** $f : A \rightarrow B$ assigns exactly one element of B to each element of A . The set A is the **domain**; B is the **codomain**.



For $f : A \rightarrow B$:

Domain	A
Codomain	B
Range	elements in B touched by f : $f[A]$

$$f[A] = \{f(a) \mid a \in A\} \subseteq B.$$

For example, let

$$f : \{1, 2, 3\} \rightarrow \{a, b, c\}, \quad f(1) = a, \quad f(2) = a, \quad f(3) = c.$$

Its range is $\{a, c\}$. The element b is in the codomain but not the range.

For $S \subseteq A$, its **image** is

$$f[S] = \{f(s) \mid s \in S\}.$$

For $T \subseteq B$, its **preimage** is

$$f^{-1}[T] = \{a \in A \mid f(a) \in T\}.$$

This preimage exists for every function; it is not an inverse function.

The **graph** of f records all input–output pairs:

$$G_f = \{(a, f(a)) \mid a \in A\} \subseteq A \times B.$$

An assignment is not a function if an input has no output or more than one output. Different inputs may share an output.

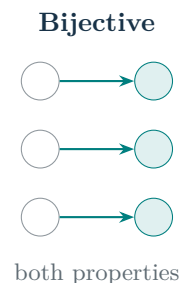
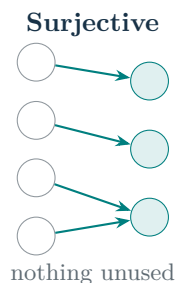
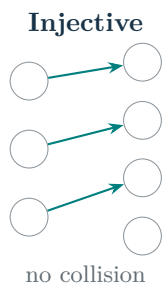
A **total function** is defined on every element of its domain. A **partial function** may be undefined for some inputs. In this course, “function” means total function unless stated otherwise.

3.2 Injection, surjection, and bijection

MUST UNDERSTAND Separate collisions from unused targets.

Let $f : A \rightarrow B$.

Property	Mathematical definition	Plain meaning
Injective	$\forall a_1, a_2 \in A, f(a_1) = f(a_2) \rightarrow a_1 = a_2$	no collisions
Surjective	$\forall b \in B, \exists a \in A, f(a) = b$	nothing unused
Bijjective	injective and surjective	exactly one input per target



The injective definition has an equivalent contrapositive form:

$$\forall a_1, a_2 \in A, \quad a_1 \neq a_2 \rightarrow f(a_1) \neq f(a_2).$$

Equal outputs force equal inputs; equivalently, distinct inputs have distinct outputs.

Squaring on the integers

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be $f(n) = n^2$.

It is not injective because $f(1) = f(-1)$. It is not surjective because, for example, -1 has no integer preimage. Its range is

$$f[\mathbb{Z}] = \{0, 1, 4, 9, \dots\}.$$

Now let

$$g : \mathbb{N} \rightarrow \{0, 1, 4, 9, \dots\}, \quad g(n) = n^2.$$

If $g(m) = g(n)$, then $m^2 = n^2$. Since $m, n \geq 0$, we get $m = n$. Every target is $k^2 = g(k)$ for some $k \in \mathbb{N}$. Thus g is bijective.

The algebraic rule did not change. The domain and codomain did.

3.3 Inverse functions

A function $f : A \rightarrow B$ has an **inverse function** $f^{-1} : B \rightarrow A$ exactly when f is bijective. It reverses every match:

$$f^{-1}(b) = a \iff f(a) = b.$$

For example, let $h : \mathbb{Z} \rightarrow \mathbb{Z}$ be $h(x) = x + 1$. Then $h^{-1}(y) = y - 1$.



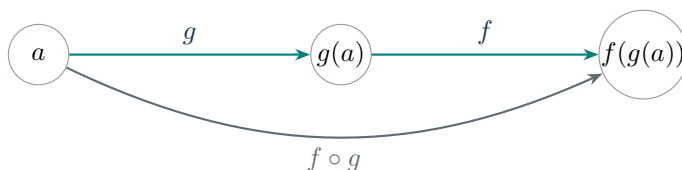
The number positions stay fixed. The inverse reverses the arrows.

3.4 Composition

If $g : A \rightarrow B$ and $f : B \rightarrow C$, their **composition** is

$$f \circ g : A \rightarrow C, \quad (f \circ g)(a) = f(g(a)).$$

Read $f \circ g$ from right to left: apply g first, then f .



For $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$,

$$(f \circ g)(x) = 2(x + 1) = 2x + 2.$$

Composition laws

Composition is **associative** whenever the compositions are defined:

$$(f \circ g) \circ h = f \circ (g \circ h).$$

The grouping changes, but the order remains h , then g , then f .

Composition is generally **not commutative**. For the functions above,

$$(f \circ g)(x) = 2x + 2, \quad (g \circ f)(x) = 2x + 1.$$

Therefore $f \circ g \neq g \circ f$.

3.5 Floor and ceiling

For $x \in \mathbb{R}$, the **floor** and **ceiling** are

$$\lfloor x \rfloor = \text{the greatest integer at most } x, \quad \lceil x \rceil = \text{the least integer at least } x.$$

For example,

$$\lfloor 2.7 \rfloor = 2, \quad \lceil 2.7 \rceil = 3, \quad \lfloor -2.7 \rfloor = -3, \quad \lceil -2.7 \rceil = -2.$$

Equivalently,

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1, \quad \lceil x \rceil - 1 < x \leq \lceil x \rceil.$$

3.6 Counting functions

Let $|A| = n$ and $|B| = m$.

For an unrestricted function $A \rightarrow B$, each of the n inputs independently chooses one of m outputs. Therefore there are

$$m^n$$

functions.

For an injective function, outputs cannot be reused. When $m \geq n$, there are

$$m(m-1) \cdots (m-n+1) = \frac{m!}{(m-n)!}$$

injections. When $m < n$, there are none. In particular, two n -element sets have $n!$ bijections between them.

3.7 Practice

Set practice

Let $A = \{1, 2\}$ and $B = \{2, 3\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, $\mathcal{P}(A)$, and $A \times B$.

Solution.

$$\begin{aligned} A \cup B &= \{1, 2, 3\}, & A \cap B &= \{2\}, & A \setminus B &= \{1\}, \\ \mathcal{P}(A) &= \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}, & A \times B &= \{(1, 2), (1, 3), (2, 2), (2, 3)\}. \end{aligned}$$

Function practice

Let $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ be $f(n) = n + 3$ and $g(n) = 2n$.

1. Find $(f \circ g)(n)$ and $(g \circ f)(n)$.
2. Decide whether each function is injective, surjective, or bijective.
3. Find the inverse of any bijective function.

Solution.

$$(f \circ g)(n) = 2n + 3, \quad (g \circ f)(n) = 2n + 6.$$

The function f is bijective, with $f^{-1}(n) = n - 3$. The function g is injective but not surjective: it reaches the even integers and misses the odd integers.

4 Cardinality: Describing the size of sets

The **cardinality** of a set describes its size. For a finite set, this is usually easy to understand: count its elements. For example,

$$|\{a, b, c, d\}| = 4.$$

Infinite sets are different. We cannot finish counting their elements one by one, and a proper subset may have the same size as the whole set. Instead, we compare sizes by asking whether the elements of two sets can be matched.

4.1 Comparing cardinalities

Sets A and B have the **same cardinality**, written $|A| = |B|$, when a bijection $A \rightarrow B$ exists. We write

$$|A| \leq |B|$$

when an injection $A \rightarrow B$ exists. Finally, $|A| < |B|$ means $|A| \leq |B|$ and no bijection $A \rightarrow B$ exists.

For infinite sets, one injective but non-surjective map does **not** prove strict inequality. The shift $s : \mathbb{N} \rightarrow \mathbb{N}$, $s(n) = n + 1$, is injective and not surjective, but the two sets are literally the same set.

For finite sets A and B with $|A| = |B|$, an injection $A \rightarrow B$ is automatically surjective, and a surjection is automatically injective. This convenient equivalence can fail for infinite sets.

The **Schroeder–Bernstein theorem** repairs a common comparison problem:

$$|A| \leq |B| \text{ and } |B| \leq |A| \implies |A| = |B|.$$

In words: injections in both directions guarantee a bijection, even when that bijection is hard to construct directly.

For example, the identity map injects $(0, 1)$ into $(0, 1]$, and $x \mapsto x/2$ injects $(0, 1]$ into $(0, 1)$. Therefore

$$|(0, 1)| = |(0, 1]|.$$

4.2 Countable sets

MUST UNDERSTAND Countable means finite or listable by $\mathbb{N}$.

A set is **countable** when it is finite or countably infinite. An infinite set S is **countably infinite** when there is a bijection

$$e : \mathbb{N} \rightarrow S.$$

Then $|S| = \aleph_0$. Equivalently, its elements can be listed as

$$e(0), e(1), e(2), \dots$$

with every element appearing exactly once.

Hilbert's hotel makes the key infinite behavior visible. Number the occupied rooms by $\mathbb{N}$.

- For one new guest, move the old guest in room n to room $n + 1$ and use room 0.
- For countably many new guests, move old guest n to room $2n$ and put new guest k in room $2k + 1$.

No guest is lost and no room is shared. A proper subset of an infinite set can therefore have the same cardinality as the full set.

Optional demo video. A visual follow-up to the hotel example: *How An Infinite Hotel Ran Out Of Room*.

The integers are countable

Define $e : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$e(0) = 0, \quad e(2k - 1) = k, \quad e(2k) = -k \quad (k \geq 1).$$

This gives the list

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Every integer occurs: 0 occurs first, positive k occurs at index $2k - 1$, and negative $-k$ occurs at index $2k$. These index formulas are distinct, so no integer repeats. Hence e is bijective and $\mathbb{Z}$ is countably infinite.

The rational numbers are countable

The rational numbers are

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Every positive rational number has a representation p/q with positive integers p and q . Arrange the pairs (p, q) by increasing $p + q$:

$$(1, 1); \quad (1, 2), (2, 1); \quad (1, 3), (2, 2), (3, 1); \quad \dots$$

Traverse each finite diagonal and list the value p/q only when it has not appeared earlier. Every positive rational appears on some diagonal, and duplicates such as $1/1 = 2/2$ are skipped. This produces a list $q_1, q_2, \dots$ of all positive rationals, with no omissions and no repetitions.

Now alternate each positive rational with its negative:

$$0, q_1, -q_1, q_2, -q_2, \dots$$

This list contains every rational number exactly once, so $\mathbb{Q}$ is countable.

4.3 Products of countable sets

Suppose the countably infinite sets A and B have lists

$$A = \{a_0, a_1, a_2, \dots\}, \quad B = \{b_0, b_1, b_2, \dots\}.$$

Every pair in $A \times B$ has the form (a_i, b_j) . List the pairs by increasing $i + j$:

$$(a_0, b_0); \quad (a_0, b_1), (a_1, b_0); \quad (a_0, b_2), (a_1, b_1), (a_2, b_0); \quad \dots$$

Each diagonal is finite, and every pair occurs on exactly one diagonal. Therefore $A \times B$ is countable. The same conclusion holds if either factor is finite. In particular,

$\mathbb{Q} \times \mathbb{Q} \text{ is countable.}$

The same diagonal idea shows that a union of finitely or countably many countable sets is countable: arrange the individual lists as rows and traverse the resulting grid diagonally, skipping repetitions.

4.4 Uncountable sets

A set is **uncountable** when it is not countable. We prove that $(0, 1)$ is uncountable by showing that every proposed list misses a number.

Suppose, for contradiction, that all numbers in $(0, 1)$ can be listed:

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}\dots \\ r_2 &= 0.d_{21}d_{22}d_{23}\dots \\ r_3 &= 0.d_{31}d_{32}d_{33}\dots \\ &\vdots \end{aligned}$$

Choose for each real its decimal expansion that does not end in an infinite string of 9s. Now construct

$$r = 0.c_1c_2c_3\dots, \quad c_i = \begin{cases} 4, & d_{ii} \neq 4, \\ 5, & d_{ii} = 4. \end{cases}$$

The number r lies in $(0, 1)$ and has no ambiguous tail of 9s. It differs from r_i in decimal position i , so $r \neq r_i$ for every i . Thus the proposed list was incomplete, a contradiction. Therefore $(0, 1)$, and hence $\mathbb{R}$, is uncountable.

The reason is not that the real line is dense. The rational numbers $\mathbb{Q}$ are also dense, yet $\mathbb{Q}$ is countable. The decisive point is the diagonal construction: it defeats every proposed list.

MUST UNDERSTAND Cantor diagonalization builds one object that differs from item i at position i .

The proof is **Cantor diagonalization**. Its reusable pattern is:

1. Assume a complete list exists.
2. Use the i th listed object to choose the i th part of a new object.
3. Ensure the new object differs from item i at position i .
4. Conclude that the list was not complete.

4.5 Products with an uncountable factor

First, the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are uncountable. Otherwise both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ would be countable, so their union $\mathbb{R}$ would be countable, contradicting Cantor's diagonal argument.

Now let U be uncountable and let C be nonempty. Fix one $c_0 \in C$. The map

$$U \longrightarrow C \times U, \quad u \longmapsto (c_0, u)$$

is injective. Thus $C \times U$ contains a copy of the uncountable set U and is itself uncountable. For example,

$$\boxed{\mathbb{Q} \times (\mathbb{R} \setminus \mathbb{Q}) \text{ is uncountable.}}$$

The order of the factors does not affect this conclusion: swapping coordinates is a bijection $A \times B \rightarrow B \times A$.

4.6 Cantor's theorem

Cantor's theorem. For every set S ,

$$|S| < |\mathcal{P}(S)|.$$

That is, the power set of S is “bigger” than S in cardinality.

The injection $x \mapsto \{x\}$ gives $|S| \leq |\mathcal{P}(S)|$. To prove the strict inequality, it remains to show that no function $f : S \rightarrow \mathcal{P}(S)$ can reach every subset in $\mathcal{P}(S)$; equivalently, no surjective f exists.

First fix any function $f : S \rightarrow \mathcal{P}(S)$. The function assigns each element $x \in S$ to one subset $f(x) \subseteq S$. Define

$$D = \{x \in S \mid x \notin f(x)\}.$$

The set D collects exactly those elements that do not belong to their own assigned subsets. It is well-defined for every f before any assumption about surjectivity, and $D \subseteq S$ gives $D \in \mathcal{P}(S)$.

For example, if $S = \{a, b, c\}$, one possible function is

$$f(a) = \{a, b\}, \quad f(b) = \{a, c\}, \quad f(c) = \{b\}.$$

For each input, ask whether that input belongs to its own assigned subset:

x	$f(x)$	$x \in f(x)?$	Put x in D ?
a	$\{a, b\}$	yes	no
b	$\{a, c\}$	no	yes
c	$\{b\}$	no	yes

Here the rule produces $D = \{b, c\}$, which is not equal to $f(a)$, $f(b)$, or $f(c)$.

Now suppose, for contradiction, that f is surjective. Because $D \subseteq S$, we have $D \in \mathcal{P}(S)$, so surjectivity gives some $d \in S$ with $f(d) = D$. At this one input,

$$d \in D \iff d \notin f(d) \iff d \notin D,$$

a contradiction. If d belongs to D , the definition says it does not; if d does not belong to D , the definition says it does. Hence f cannot be surjective, so no bijection $S \rightarrow \mathcal{P}(S)$ exists.

There is no circular reasoning. The diagonal set is constructed from an arbitrary f . Surjectivity is used only afterward, to claim that some input d must map to that already-defined set D .

This theorem shows that there is no largest infinity: taking a power set always produces a strictly larger cardinality.

4.7 The continuum hypothesis

The size of the real numbers is the **continuum**:

$$\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}.$$

A subset of $\mathbb{N}$ corresponds to an infinite bit string. Such strings have the same cardinality as $\mathbb{R}$; a standard encoding handles the two binary expansions of dyadic rationals. Cantor's theorem gives

$$\aleph_0 < 2^{\aleph_0} = \mathfrak{c}.$$

The symbol $\aleph_1$ denotes the smallest uncountable cardinality. Thus

$$\aleph_0 < \aleph_1 \leq \mathfrak{c}.$$

The **continuum hypothesis** (CH) asks whether the continuum is the smallest uncountable cardinality:

$$\boxed{\mathfrak{c} = \aleph_1?}$$

This is stronger than saying that $\mathbb{R}$ is uncountable.

4.8 Independence from ZFC

This is optional. **ZFC** is Zermelo–Fraenkel set theory plus the Axiom of Choice, the standard axiom system for most mathematics. It is **consistent** if its axioms do not produce a contradiction.

A statement is **independent of ZFC** if ZFC proves neither it nor its negation. Assuming ZFC is consistent, both

$$\text{ZFC} + \text{CH} \quad \text{and} \quad \text{ZFC} + \neg\text{CH}$$

are consistent. Gödel established the first direction in 1940; Cohen established the second in 1963. ZFC alone therefore cannot decide CH.

4.9 Cardinality and computation

A function is **computable** if a finite program produces its value on every allowed input. The set of finite programs is countable, so there are only countably many computable functions.

On the other hand, a function $b : \mathbb{N} \rightarrow \{0, 1\}$ is exactly the membership indicator of a subset of $\mathbb{N}$:

$$S_b = \{n \in \mathbb{N} \mid b(n) = 1\}.$$

These functions are in bijection with $\mathcal{P}(\mathbb{N})$ and are uncountable. Therefore some functions are uncomputable. This counting argument proves existence; it does not identify a specific uncomputable function.

5 Review exercises

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^2 + 1$. Determine its range and decide whether it is injective or surjective. Then choose a domain and codomain that make the same rule bijective.
2. How many functions are there from a 4-element set to a 3-element set? How many are injective?
3. The even natural numbers form a proper subset of $\mathbb{N}$. Must they have smaller cardinality?
4. Explain in one sentence why the diagonal real r cannot equal the i th number in the proposed list.

Short answers

1. The range is $[1, \infty)$; the stated function is neither injective nor surjective. Restrict to $[0, \infty) \rightarrow [1, \infty)$ for a bijection.
2. $3^4 = 81$ functions and no injections.
3. No. The bijection $n \mapsto 2n$ shows that the even natural numbers and $\mathbb{N}$ have the same cardinality.
4. Its i th decimal digit was chosen to differ from the i th digit of r_i .

6 Summary

- Sets are determined by membership; their operations follow the corresponding logical operations.
- A function assigns exactly one output to each input. Injection, surjection, and bijection describe how inputs meet targets.
- Functions compare cardinalities. For infinite sets, a proper subset may have the same size as the full set.
- Countable sets can be listed. Diagonalization proves that the real numbers cannot be listed.
- Cantor's theorem gives $|S| < |\mathcal{P}(S)|$; the continuum hypothesis asks whether $2^{\aleph_0} = \aleph_1$.

The next topic is basic counting: systematic methods for counting possible outcomes.